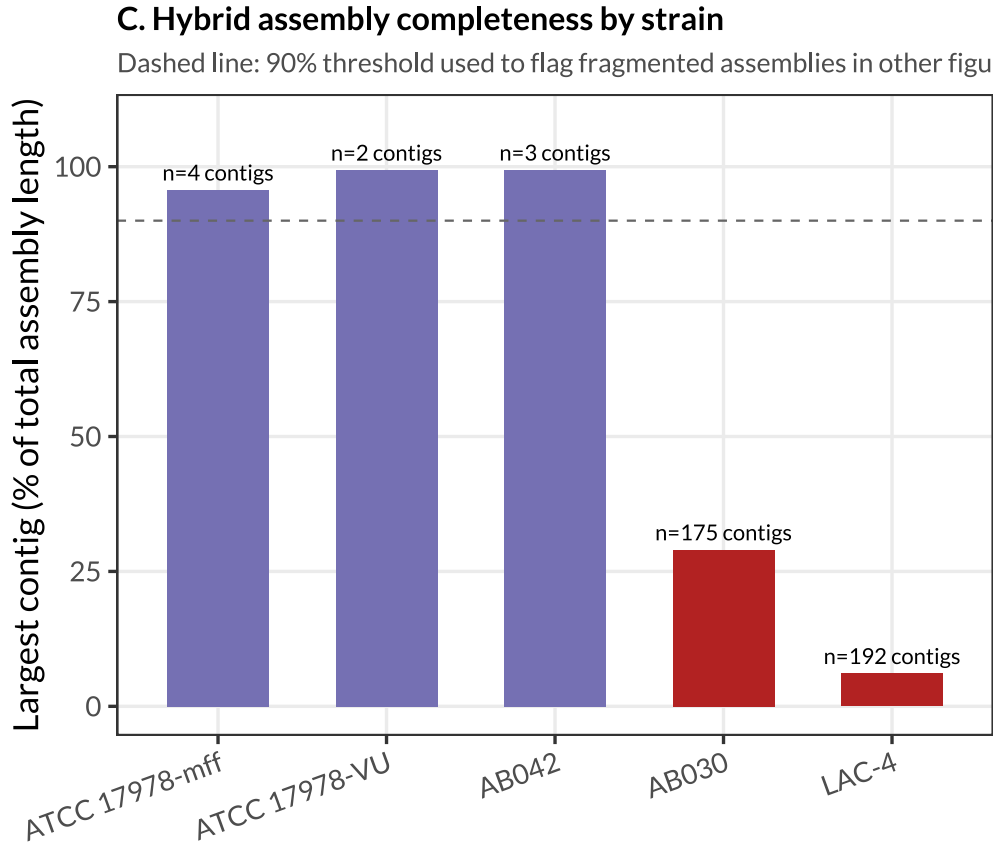
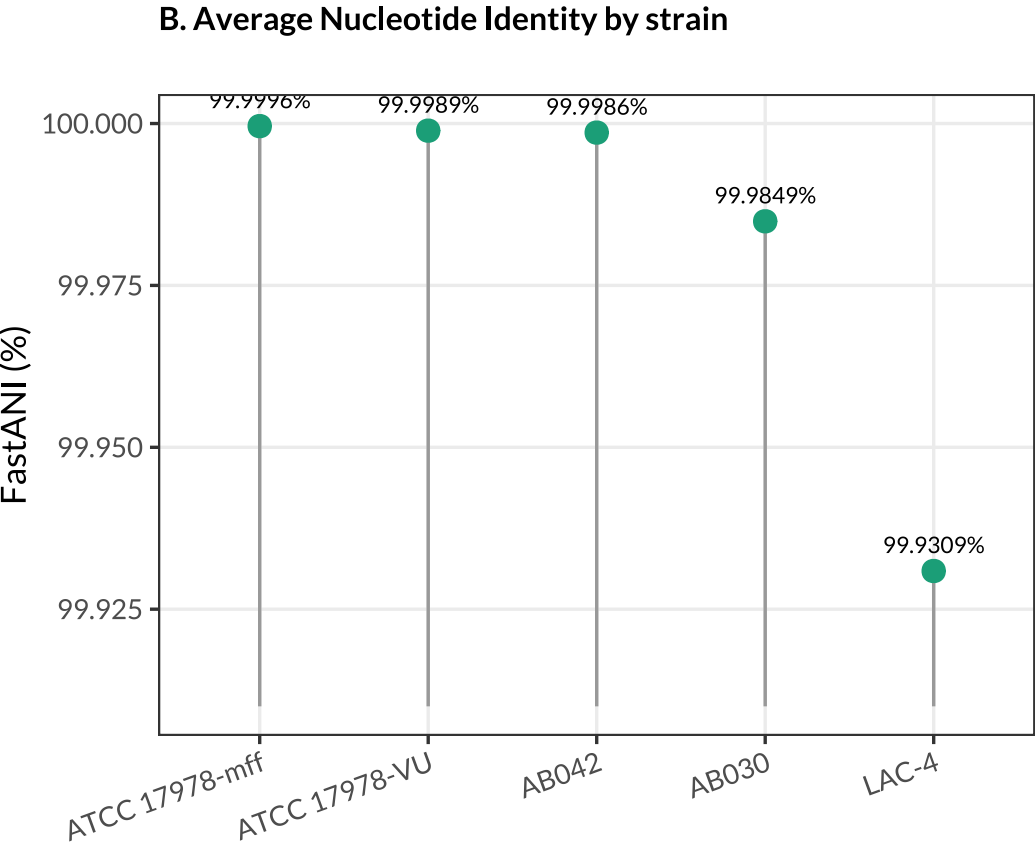
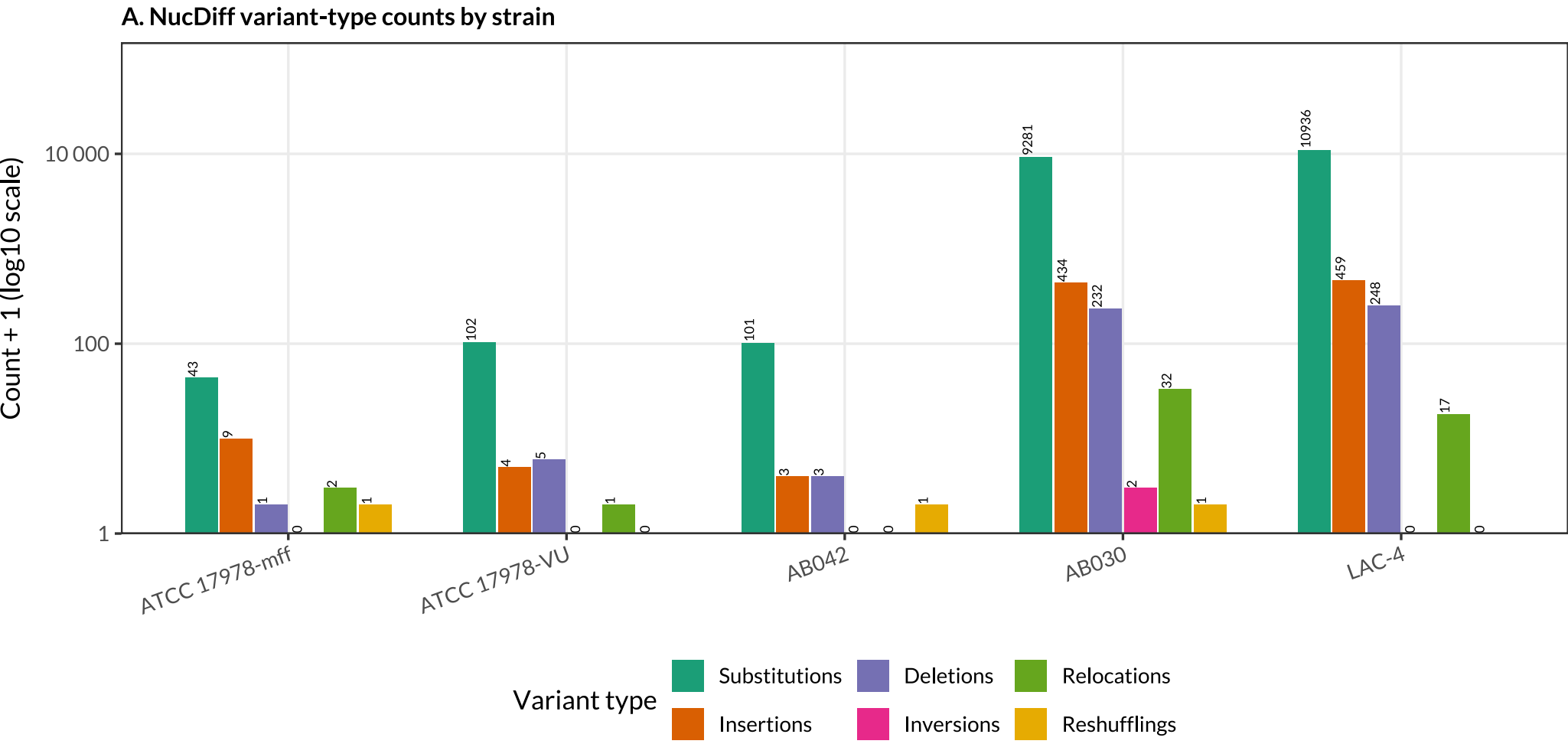


Hybrid Assembly vs. Reference Genome: Cross-Strain Comparison Summary



Cross-strain summary of hybrid assembly (Unicycler) vs. public reference genome comparisons (FastANI + NucDiff), all 5 strains. Panel A: NucDiff variant-type counts (translocations omitted, 0 in all strains; reshuffling counts reflect whole-genome start-point offsets between circular sequences, not confirmed internal rearrangement -- see per-strain dot plots/ideograms). Panel B: FastANI average nucleotide identity. Panel C: assembly completeness (largest contig as % of total assembly length); AB030 and Lac4 fall well below the 90% completeness threshold, meaning their elevated variant counts in Panel A are substantially assembly-fragmentation artifacts rather than confirmed biological variation.